

Homework for Week 12

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1.

$$(1) \quad \frac{d}{dx} \left(x^2 \frac{dy}{dx} \right) + (\lambda x + x^2) = 0, p(x) = x^2, q(x) = -x^2, w(x) = x.$$

$$(2) \quad \frac{d}{dx} \left[x^a (x-1)^{b-a} \frac{dy}{dx} \right] + \lambda x^{a-1} (x-1)^{b-a-1} y = 0,$$

$$p(x) = x^a (x-1)^{b-a}, q(x) = 0, w(x) = x^{a-1} (x-1)^{b-a-1}.$$

$$(3) \quad \frac{d}{dx} \left[\frac{x e^{-x}}{(x-1)^2} \frac{dy}{dx} \right] + \frac{\lambda e^{-x}}{(x-1)^2} y = 0, p(x) = \frac{x e^{-x}}{(x-1)^2}, q(x) = 0, w(x) = \frac{e^{-x}}{(x-1)^2}.$$

$$(4) \quad \frac{d}{dx} \left(e^{-x^2} \frac{dy}{dx} \right) + 2\lambda e^{-x^2} = 0, p(x) = e^{-x^2}, q(x) = 0, w(x) = 2e^{-x^2}.$$

2. 考虑 $y(x) \in D(\hat{L}), z(x) \in L^2[a, b]$, 有

$$\begin{aligned} \langle z | \hat{L} y \rangle &= \langle \hat{L}^\dagger z | y \rangle \\ \Rightarrow \langle z | \hat{L} y \rangle &= \int_a^b z^* \left\{ -\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x) y \right\} dx \\ &= -z^* p(x) \frac{dy}{dx} \Big|_a^b + \int_a^b p(x) \frac{dz^*}{dx} \frac{dy}{dx} dx + \int_a^b q(x) z^*(x) y(x) dx \\ &= \left[y(x) p(x) \frac{dz^*(x)}{dx} - z^*(x) p(x) \frac{dy(x)}{dx} \right]_a^b - \int_a^b \frac{d}{dx} \left[p(x) \frac{dz^*}{dx} \right] y(x) dx + \int_a^b q(x) z^*(x) y(x) dx \\ &= \left[y(x) p(x) \frac{dz^*(x)}{dx} - z^*(x) p(x) \frac{dy(x)}{dx} \right]_a^b + \langle \hat{L}' z | y \rangle = \langle \hat{L}^\dagger z | y \rangle. \end{aligned}$$

若 $\hat{L}$ 的伴算符存在, 即要求边界项为 0, 整理并代入 $y \in D(\hat{L})$ 满足的边界条件得到

$$y(a) \left[\frac{\cos \alpha}{c_1} z'^*(b) - \frac{\cos \beta}{c_2} z^*(b) - z'^*(a) \right] + y'(a) \left[-\frac{\sin \alpha}{c_1} z'^*(b) - \frac{\sin \beta}{c_2} z^*(b) + z^*(a) \right] = 0.$$

上式对任意 $y(a), y'(a)$ 成立，故伴算符存在条件为

$$z'(a) = \frac{\cos \alpha}{c_1} z'(b) - \frac{\cos \beta}{c_2} z(b), \quad (0.1)$$

$$z(a) = \frac{\sin \alpha}{c_1} z'(b) + \frac{\sin \beta}{c_2} z(b), \quad (0.2)$$

此时 $\hat{L}$ 的伴算符为

$$\hat{L}^\dagger = -\frac{d}{dx} \left[p(x) \frac{d}{dx} \right] + q(x),$$

$$D(\hat{L}) = \{y(x) : a \leq x \leq b, y(x) \in L^2[a, b] \text{ 且满足 (0.1) 和 (0.2)}\},$$

当且仅当 $\sin(\alpha + \beta) = c_1 c_2$ 时 $\hat{L}$ 是自伴算符.

3. 本征函数具有形式 $e^{\pm \lambda x}$ ，取线形组合并带入边界条件得到

$$C_1 + C_2 = 0,$$

$$(1 - \beta)C_1 e^{i\lambda l} - (1 + \beta)C_2 e^{-i\lambda l} = 0.$$

由以上两式得到

$$2i\lambda l = \ln \frac{\beta + 1}{\beta - 1},$$

并考虑到对数函数的多值性与 $0 < \beta \leq 1$ ，取近似得到本征值

$$2i\lambda l = \ln \frac{\beta + 1}{\beta - 1} \approx 2\beta + i(2n + 1)\pi \Rightarrow \lambda_n = \frac{-2i\beta + (2n + 1)\pi}{2l}, n \in \mathbb{Z}.$$

相应的本征函数为 $\{e^{\beta x/l} e^{i \frac{(2n+1)x}{2l}} - e^{-\beta x/l} e^{-i \frac{(2n+1)x}{2l}}\}$.

4. 本征函数具有形式 $e^{-i\lambda x}$ ，带入边界条件得到

$$e^{-i\lambda b} = e^{i\theta} e^{-i\lambda a} \Rightarrow \lambda_n = \frac{\theta + 2n\pi}{a - b}, n \in \mathbb{Z},$$

相应的本征函数为 $\{e^{i \frac{\theta + 2n\pi}{b-a} x}\}$.